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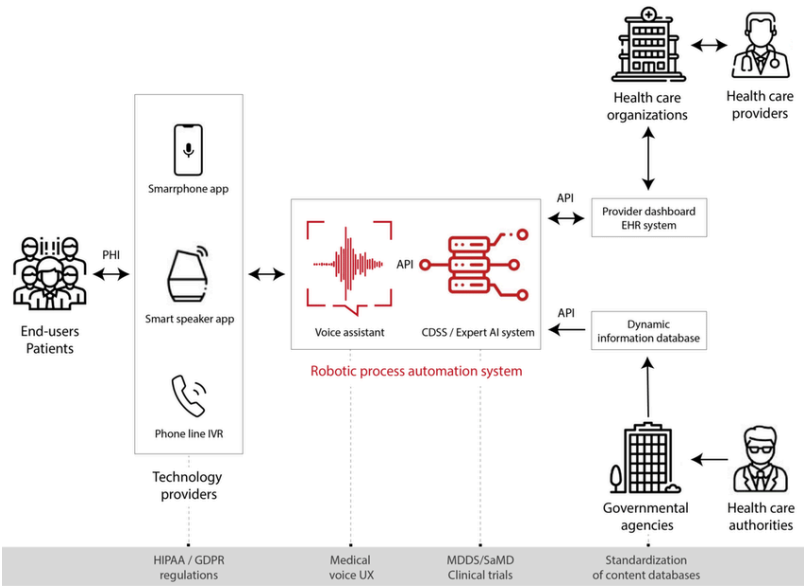
Artificial Intelligence in Public Health: Advantages of AI-Based Diagnostic Systems in Reducing Healthcare Access Gaps in the United States

Artificial Intelligence has become an increasingly relevant tool in government and public sector services, particularly in the field of public health. In the United States, where access to healthcare is often shaped by economic inequality, geographic barriers, and insurance limitations, AI-based diagnostic systems are emerging as an accessible alternative for individuals seeking medical guidance. These systems allow users to describe symptoms and receive immediate feedback, which positions AI as an initial point of contact within the healthcare process. From this perspective, the integration of AI into public health reflects not only technological advancement but also a response to structural gaps in access to care (Pew Research Center, 2023; World Health Organization, 2023).

The need for such tools becomes evident when considering the characteristics of the U.S. healthcare system. Access to timely diagnosis is not equally distributed, as individuals with stable income and insurance coverage are more likely to receive preventive care and specialist attention, while underserved populations often delay or avoid seeking care due to cost and availability constraints. Consequently, AI diagnostic tools address a critical gap by offering immediate and low-cost access to health-related information. In this context, AI does not replace healthcare services but rather functions as a complementary resource that supports individuals in navigating uncertainty (U.S. Department of Health and Human Services, 2024; OECD, 2023).

AI diagnostic systems are typically built on machine learning models and natural language processing techniques (Figure 2), which enable them to analyze large volumes of data and generate responses based on patterns. When a user inputs symptoms, the system compares this information with existing datasets and produces possible explanations or recommendations. This process, although not equivalent to clinical evaluation, provides users with a structured interpretation of their concerns. Therefore, one of the main advantages of AI in this context is its ability to translate complex medical information into accessible and understandable responses (Topol, 2023).

Figure 1. AI Diagnostic Process and Its Role in Improving Healthcare Access



A key advantage of AI diagnostic systems is their capacity to improve accessibility. Unlike traditional healthcare services, which often require appointments, transportation, and financial resources, AI tools are available at any time and can be accessed from virtually any location. This is particularly significant for individuals living in rural areas or communities with limited healthcare infrastructure. In these cases, AI serves as a bridge between the individual and the healthcare system, reducing the initial barrier to entry. As a result, more people can engage with their health concerns at an earlier stage (OECD, 2023).

To analyze this phenomenon, the study is based on existing public health and digital access literature (OECD, 2023, 2025; Pew Research Center, 2023; U.S. Department of Health and Human Services, 2024). In the table 1, the analysis compares access across three population groups: urban residents, rural residents, and low-income populations, which represent different levels of structural inequality in the United States. In this context, traditional access refers to the ability to obtain healthcare through conventional systems, including in-person medical visits, insurance-based services, and physical healthcare infrastructure. In contrast, AI-based digital access refers to the use of artificial intelligence tools and digital health platforms that provide immediate, remote, and low-cost health information. Therefore, this comparison allows for a clearer understanding of how AI can improve access to healthcare by reducing geographic, economic, and systemic barriers.

Table 1. Comparison of Traditional Healthcare Access and AI-Based Digital Access Across Population Groups in the United States

Group	**Traditional Access (%)**	**AI/Digital Access (%)**
Urban	70	82
Rural	40	65
Low-Income	35	60

However, while the advantages of AI are significant, it is also necessary to recognize certain limitations in order to understand its role more accurately. AI systems lack clinical judgment and do not have access to individual medical histories, so their responses are based on generalized patterns rather than personalized evaluation. In addition, these systems often fail to account for contextual factors such as socioeconomic status, cultural background, or access to healthcare resources. For example, a recommendation to consult a specialist may be appropriate from a clinical perspective but unrealistic for individuals facing financial or logistical barriers (Topol, 2023; World Health Organization, 2023).

Despite these limitations, the advantages of AI diagnostic systems remain relevant, particularly when they are understood as complementary tools rather than replacements for professional care. In fact, their value becomes more evident when integrated into broader public health strategies. For this reason, an innovative application of AI would involve developing publicly regulated diagnostic assistants integrated with healthcare systems. These tools could maintain the advantages of accessibility and efficiency while improving the relevance of their recommendations by incorporating contextual information (Floridi et al., 2022).

Such systems could, for example, guide users toward low-cost clinics, public health programs, or telehealth services based on their location and available resources. Additionally, they could include risk indicators and clear recommendations to seek professional care when necessary. This approach would strengthen AI's role as a supportive tool that not only provides information but also facilitates more realistic access to services. As a result, AI would contribute more directly to reducing healthcare inequalities (U.S. Department of Health and Human Services, 2024).

In conclusion, AI-based diagnostic systems offer clear advantages in the public health sector, particularly in improving accessibility, efficiency, scalability, and health awareness. In the United States, where access to healthcare remains uneven, these tools play an important role in reducing initial barriers and supporting individuals in navigating health-related decisions. While limitations exist, they do not negate the value of AI; rather, they highlight the importance of responsible implementation. Ultimately, the integration of AI into public health should focus on maximizing its benefits while ensuring it aligns with the complexity of human needs and social realities (World Health Organization, 2023).

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